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Container with a movable separation panel

Abstract

A container movable between a flat configuration and a box configuration is disclosed. The container comprises a first side panel hingedly coupled to a first side portion, a second side panel hingedly coupled to a second side portion, a separation panel, and a bottom panel. The separation panel is hingedly coupled to the first side portion and to the second side portion. When the container is in the box configuration, the bottom panel and the separation panel face each other, and the first side panel and the second side panel face each other. The container is further movable between the box configuration and a formed configuration. When the container is in the formed configuration, the first side portion and the first side panel face each other, the second side portion and the second side panel face each other, and the separation panel and the bottom panel are substantially orthogonal.

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References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
2785846	12/1956	Weiner	229/117.22	B65D 5/46096
3158286	12/1963	Phillips, Jr.	229/117	B65D 71/0022
3283949	12/1965	Wysocki	206/187	B65D 71/0022
3570706	12/1970	Harrelson	206/187	B65D 71/0022
3780906	12/1972	Katzenmeyer	N/A	N/A
4256223	12/1980	Pawlowski	206/192	B65D 5/4204
4782943	12/1987	Blackman	206/175	B65D 71/0022
5775574	12/1997	Whitnell	206/175	B65D 5/46184
5816411	12/1997	Smith	206/769	B65D 5/46096
7383949	12/2007	Cuomo	N/A	N/A
7438181	12/2007	Cuomo	N/A	N/A
8297437	12/2011	Smalley	N/A	N/A
8622207	12/2013	Smalley	N/A	N/A
9415914	12/2015	Holley, Jr.	N/A	N/A
2006/0091190	12/2005	Nikolai	229/117.22	B65D 5/067
2013/0202225	12/2012	Metcalfe	N/A	N/A
2021/0024247	12/2020	Sollie	N/A	B65D 5/4266
2021/0292042	12/2020	Sollie	N/A	B65D 5/4266

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
1806349	12/1969	DE	N/A
2242675	12/1990	GB	B65D 71/30
6077588	12/2016	JP	N/A
20140066046	12/2013	KR	N/A
WO-2006050207	12/2005	WO	B65D 5/064

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) The present application claims priority to U.S. 63/451,345, filed Mar. 10, 2023, the contents of which are incorporated by reference herein in its entirety.

TECHNICAL FIELD
(1) Embodiments are in the field of containers. More particularly, embodiments disclosed herein relate to containers which, inter alia, foster a container with a movable separation panel. The container can be rapidly and easily formed. The container may be used to carry cups and beverages. The container provides convenience while reducing material and manufacturing costs.
BACKGROUND
(2) There are several challenges that are associated with packaging food, cups, beverages, consumer products, etc. For example, containers may be required to be held for a variety of purposes such as carrying, lifting, pouring, serving, using, delivering, catering, storing, moving, transporting, etc. In addition, these containers may be required to be formed quickly and conveniently.
(3) To address the previous challenges, a number of packaging solutions have been proposed. These packaging solutions (e.g., extended handles, cup carriers, large folding trays, etc.) use significantly more material than required by typical containers. As a result, cost effectiveness may be compromised. In addition, these packaging solutions may not provide a convenient box forming experience.
(4) Thus, it is desirable to provide a container that is able to overcome the above disadvantages.
(5) Advantages of the present disclosure will become more fully apparent from the detailed description.
BRIEF SUMMARY
(6) Embodiments of the present disclosure are drawn to a container, where the container or a blank used to make the container comprises two side panels, two side portions, a separation panel, and a bottom panel.
(7) A first aspect of the present disclosure is drawn to embodiments directed to a container and to blanks used for making the container. The container is movable between a flat configuration and a box configuration. The container comprises a first side panel hingedly coupled to a first side portion, a second side panel hingedly coupled to a second side portion, a separation panel, and a bottom panel. The separation panel is hingedly coupled to the first side portion. The separation panel is hingedly coupled to the second side portion. The separation panel comprises a handle. The bottom panel and the separation panel face each other when the container is in the box configuration. The first side panel and the second side panel face each other when the container is in the box configuration. The container is further movable between the box configuration and a formed configuration. The first side portion is configured to fold towards the first side panel. The second side portion is configured to fold towards the second side panel. The first side portion and the first side panel face each other when the container is in the formed configuration. The second side portion and the second side panel face each other when the container is in the formed configuration. The separation panel and the bottom panel are substantially orthogonal with respect to each other when the container is in the formed configuration. The container is configured to be held via the handle when the container is in the formed configuration.
(8) A second aspect of the present disclosure is drawn to embodiments directed to a container and to blanks used for making the container. The container is movable between a flat configuration and

a box configuration. The container comprises a first side panel hingedly coupled to a first side portion, a second side panel hingedly coupled to a second side portion, a separation panel, and a bottom panel. The separation panel is hingedly coupled to the first side portion. The separation panel is hingedly coupled to the second side portion. The bottom panel and the separation panel face each other when the container is in the box configuration. The first side panel and the second side panel face each other when the container is in the box configuration. The container is further movable between the box configuration and a formed configuration. The first side portion is configured to fold towards the first side panel. The second side portion is configured to fold towards the second side panel. The first side portion and the first side panel face each other when the container is in the formed configuration. The second side portion and the second side panel face each other when the container is in the formed configuration. The separation panel and the bottom panel are substantially orthogonal with respect to each other when the container is in the formed configuration.
(9) Additional embodiments and additional features of embodiments for containers or for blanks used for making containers are described below and are hereby incorporated into this section.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) The detailed description will refer to the following drawings, wherein like reference numerals refer to like elements, and wherein:
(2) FIG. **1**A is a diagram illustrating a plan view of an embodiment of a blank used for making a container. The blank comprises a first side panel hingedly coupled to a first side portion, a second side panel hingedly coupled to a second side portion, a separation panel, and a bottom panel. The separation panel is hingedly coupled to the first side portion. The separation panel is hingedly coupled to the second side portion. The separation panel comprises a handle. The blank also comprises two end flaps and a complementary side panel;
(3) FIG. **1**B is a diagram illustrating the blank shown in FIG. **1**A, wherein the separation panel is folded towards the first side portion;
(4) FIG. **1**C is a diagram illustrating the blank shown in FIG. **1**A. In addition to the folding shown in FIG. **1**B, the two end flaps are folded towards the first side panel;
(5) FIG. **1**D is a diagram illustrating a top view of a flat configuration of the container made from the blank shown in FIG. **1**A. In addition to the folding shown in FIG. **1**C, the complementary side panel is folded towards the bottom panel;
(6) FIG. **1**E is a diagram illustrating a perspective view of a box configuration of the container made from the blank shown in FIG. **1**A;
(7) FIG. **1**F is a diagram illustrating a perspective view of a formed configuration of the container made from the blank shown in FIG. **1**A;
(8) FIG. **2**A is a diagram illustrating a plan view of an embodiment of a blank used for making a container. The blank comprises a first side panel hingedly coupled to a first side portion, a second side panel hingedly coupled to a second side portion, a third side panel hingedly coupled to a third side portion, a fourth side panel hingedly coupled to a fourth side portion, a separation panel (also referred to as a first separation panel), a second separation panel, and a bottom panel. The separation panel is hingedly coupled to the first side portion. The separation panel is hingedly coupled to the second side portion. The separation panel comprises a handle. The second separation panel is hingedly coupled to the third side portion. The second separation panel is hingedly coupled to the fourth side portion. The blank also comprises two end flaps;
(9) FIG. **2**B is a diagram illustrating the blank shown in FIG. **2**A, wherein the separation panel is folded towards the first side portion;

(10) FIG. **2**C is a diagram illustrating the blank shown in FIG. **2**A. In addition to the folding shown in FIG. **2**B, the two end flaps are folded towards the first side panel and the fourth side portion is folded towards the second separation panel;
(11) FIG. **2**D is a diagram illustrating a top view of a flat configuration of the container made from the blank shown in FIG. **2**A. In addition to the folding shown in FIG. **2**C, the third side panel is folded towards the bottom panel;
(12) FIG. **2**E is a diagram illustrating a perspective view of a box configuration of the container made from the blank shown in FIG. **2**A;
(13) FIG. **2**F is a diagram illustrating a perspective view of a formed configuration of the container made from the blank shown in FIG. **2**A; and
(14) FIGS. **3**A-**3**F, **4**A-**4**F, and **5**A-**5**B are diagrams illustrating plan views of embodiments of blanks used for making containers. The blanks shown in FIGS. **3**A-**3**F and in FIGS. **5**A-**5**B are alternative configurations to the blank shown in FIG. **1**A. The blanks shown in FIGS. **4**A-**4**F are alternative configurations to the blank shown in FIG. **2**A.

DETAILED DESCRIPTION

(15) It is to be understood that the figures and descriptions of the present disclosure may have been simplified to illustrate elements that are relevant for a clear understanding of the present disclosure, while eliminating, for purposes of clarity, other elements found in a typical container or typical method of using a container. Those of ordinary skill in the art will recognize that other elements may be desirable and/or required in order to implement the embodiments disclosed herein. However, because such elements are well known in the art, and because they do not facilitate a better understanding of the present disclosure, a discussion of such elements is not provided herein. It is also to be understood that the drawings included herewith only provide diagrammatic representations of the presently preferred structures of the present disclosure and that structures falling within the scope of the present disclosure may include structures different than those shown in the drawings. Reference will now be made to the drawings wherein like structures are provided with like reference designations.
(16) For purposes of this disclosure, the term “planar” refers to an element or combination of elements that may have any thickness, and having sides defining the thickness that are parallel with each other.
(17) For purposes of this disclosure, the expression “A is hingedly coupled to B” refers to an element A being movably coupled to an element B via a hinge. A hinge may comprise a perforation, a crease, a score, a bend, a section with less thickness than surrounding material, a section with less density than surrounding material, and a combination thereof. The coupling of element A and element B via a hinge may require folding, joining, and/or attaching elements of a blank (e.g. by using a glue tab, a glue flap, an extended tab, an extended flap, etc.)
(18) For purposes of this disclosure, the expression “A is configured to fold towards B” refers to at least a portion of element A being configured to fold towards at least a portion of element B via a hinge.
(19) For purposes of this disclosure, the “usable space of the container” refers to a space that is configured to contain content to be carried and/or to be held in the container when the container is in a formed configuration.
(20) For purposes of this disclosure, the expression “A and B are substantially orthogonal with respect to each other” refers to the angle defined by element A and element B being within a reasonable range from a right angle (e.g., 90°±30°).
(21) Embodiments are directed to a container. Embodiments are also directed to a blank used for making a container. The container may be used for packaging various products (e.g., cups, beverages, bottles, consumer products, etc.), and for shipping (e.g., moving, carrying, deliveries, catering, etc.), or other suitable uses. The uses of the container are abundant. The spectrum of users may include manufacturers, restaurants, retailers, stores, distributors (e.g., fulfillment centers,

warehouses, storage facilities, etc.), end-customers, and consumers. For example, the container may be used to carry cups, beverages and bottles for take-out, catering, food delivery, food transportation, etc.

(22) The composition of the containers may vary. Portions or all of the container may comprise a suitable flexible material that allows for folding, collapsing, and moving such as cardboard, paperboard, paper, corrugated carton, plastic, corrugated plastic, polymers, metal, or combinations thereof. In addition, the container may comprise a plurality of different suitable materials. The material of any portion of the container may be chosen for reusability of the container, or, alternatively, the material may be chosen based on a disposable (i.e., a one-time or limited use) variation.

(23) The size, dimensions, thickness, shape, and weight of the container or portions of the container may vary. Portions or all of the container may be configured based on the desired overall features of the container, content to be contained in the container, and/or content to be carried in the container. Hence, the size, dimensions, thickness, shape and weight of portions or all of the container may vary. Moreover, the position, the length and the orientation of the various hinges may be configured based on the desired overall features of the container.

(24) The manufacturing of the container may vary. Desired objectives for the manufacturing processes may include reducing material costs, reducing manufacturing costs, reducing material waste, reducing manufacturing complexity, reducing shipping costs, or a combination thereof. The container may be made from a single sheet of material or may be assembled by coupling (e.g., connecting, attaching, joining, etc.) various portions of a plurality of independent sheets of material. Furthermore, portions or all of the container may comprise more than one layer of material. For example, multiple layers of material may be attached or glued to each other.

(25) The manufacturing of the blank may vary. Desired objectives for the manufacturing processes may include reducing material costs, reducing manufacturing costs, reducing material waste, reducing manufacturing complexity, reducing shipping costs, or a combination thereof. The blank may be made from a single sheet of material or may be assembled by coupling (e.g., connecting, attaching, joining, etc.) various portions of a plurality of independent sheets of material.

(26) FIG. **1**A is a diagram illustrating a plan view of an embodiment of a blank **100** used for making a container. The blank comprises a first side panel **101** hingedly coupled to a first side portion **102**, a second side panel **111** hingedly coupled to a second side portion **112**, a separation panel **103**, and a bottom panel **107**. The separation panel **103** is hingedly coupled to the first side portion **102**. The separation panel **103** is hingedly coupled to the second side portion **112**. The separation panel **103** comprises a handle **104**. The blank **100** also comprises two end flaps **115**, **125** and a complementary side panel **121**.

(27) FIG. **1**B is a diagram illustrating the blank **100** shown in FIG. **1**A. The separation panel **103** is folded towards the first side portion **102**.

(28) FIG. **1**C is a diagram illustrating the blank **100** shown in FIG. **1**A. In addition to the folding shown in FIG. **1**B, the two end flaps **115**, **125** are folded towards the first side panel **101**.

(29) FIG. **1**D is a diagram illustrating a top view of a flat configuration of the container made from the blank **100** shown in FIG. **1**A. In addition to the folding shown in FIG. **1**C, the complementary side panel **121** is folded towards the bottom panel **107**.

(30) FIG. **1**E is a diagram illustrating a perspective view of a box configuration of the container made from the blank **100** shown in FIG. **1**A.

(31) FIG. **1**F is a diagram illustrating a perspective view of a formed configuration of the container made from the blank **100** shown in FIG. **1**A.

(32) FIG. **2**A is a diagram illustrating a plan view of an embodiment of a blank **200** used for making a container. The blank **200** comprises a first side panel **201** hingedly coupled to a first side portion **202**, a second side panel **211** hingedly coupled to a second side portion **212**, a third side panel **221** hingedly coupled to a third side portion **222**, a fourth side panel **231** hingedly coupled to

a fourth side portion **232**, a separation panel **203** (also referred to as a first separation panel **203**), a second separation panel **213**, and a bottom panel **207**. The separation panel **203** is hingedly coupled to the first side portion **202**. The separation panel **203** is hingedly coupled to the second side portion **212**. The separation panel **203** comprises a handle **204**. The second separation panel **213** is hingedly coupled to the third side portion **222**. The second separation panel **213** is hingedly coupled to the fourth side portion **232**. The blank **200** also comprises two end flaps **215**, **225**.
(33) FIG. **2**B is a diagram illustrating the blank **200** shown in FIG. **2**A. The separation panel **203** is folded towards the first side portion **202**.
(34) FIG. **2**C is a diagram illustrating the blank **200** shown in FIG. **2**A. In addition to the folding shown in FIG. **2**B, the two end flaps **215**, **225** are folded towards the first side panel **201** and the fourth side portion **232** is folded towards the second separation panel **213**.
(35) FIG. **2**D is a diagram illustrating a top view of a flat configuration of the container made from the blank **200** shown in FIG. **2**A. In addition to the folding shown in FIG. **2**C, the third side panel **221** is folded towards the bottom panel **207**.
(36) FIG. **2**E is a diagram illustrating a perspective view of a box configuration of the container made from the blank **200** shown in FIG. **2**A.
(37) FIG. **2**F is a diagram illustrating a perspective view of a formed configuration of the container made from the blank **200** shown in FIG. **2**A.
(38) FIGS. **3**A-**3**F, **4**A-**4**F, and **5**A-**5**B are diagrams illustrating plan views of embodiments of blanks **300***a*, **300***b*, **300***c*, **300***d*, **300***e*, **300***f*, **400***a*, **400***b*, **400***c*, **400***d*, **400***e*, **400***f*, **500***a*, **500***b* used for making containers. The blanks **300***a*, **300***b*, **300***c*, **300***d*, **300***e*, **300***f*, **500***a*, **500***b* shown in FIGS. **3**A-**3**F and in FIGS. **5**A-**5**B are alternative configurations to the blank **100** shown in FIG. **1**A. For example, the second side panel **311** shown in FIG. **3**E is extended in comparison to the second side panel **111** shown in FIG. **1**A. The blanks **400***a*, **400***b*, **400***c*, **400***d*, **400***e*, **400***f* shown in FIGS. **4**A-**4**F are alternative configurations to the blank **200** shown in FIG. **2**A. For example, the second side panel **411** shown in FIG. **4**A is not extended in comparison to the second side panel **211** shown in FIG. **2**A.
(39) With reference to FIGS. **1**D-IF and **2**D-**2**F, embodiments are directed to a container movable between a flat configuration (e.g., FIGS. **1**D and **2**D) and a box configuration (e.g., FIGS. **1**E and **2**E). The container comprises a first side panel **101**, **201** hingedly coupled to a first side portion **102**, **202**, a second side panel **111**, **211** hingedly coupled to a second side portion **112**, **212**, a separation panel **103**, **203**, and a bottom panel **107**, **207**. The separation panel **103**, **203** is hingedly coupled to the first side portion **102**, **202**. The separation panel **103**, **203** is hingedly coupled to the second side portion **112**, **212**. The separation panel **103**, **203** comprises a handle **104**, **204**. The bottom panel **107**, **207** and the separation panel **103**, **203** face each other when the container is in the box configuration (e.g., FIGS. **1**E and **2**E). The first side panel **101**, **201** and the second side panel **111**, **211** face each other when the container is in the box configuration (e.g., FIGS. **1**E and **2**E). The container is further movable between the box configuration (e.g., FIGS. **1**E and **2**E) and a formed configuration (e.g., FIGS. **1**F and **2**F). The first side panel **101**, **201** and the second side panel **111**, **211** face each other when the container is in the formed configuration (e.g., FIGS. **1**F and **2**F). The first side portion **102**, **202** is configured to fold towards the first side panel **101**, **201**. The second side portion **112**, **212** is configured to fold towards the second side panel **111**, **211**. The first side portion **102**, **202** and the first side panel **101**, **201** face each other when the container is in the formed configuration (e.g., FIGS. **1**F and **2**F; the first side portion **102**, **202** is not visible). The second side portion **112**, **212** and the second side panel **111**, **211** face each other when the container is in the formed configuration (e.g., FIGS. **1**F and **2**F). The separation panel **103**, **203** and the bottom panel **107**, **207** are substantially orthogonal with respect to each other when the container is in the formed configuration (e.g., FIGS. **1**F and **2**F). The container is configured to be held via the handle **104**, **204** when the container is in the formed configuration (e.g., FIGS. **1**F and **2**F).
(40) In an embodiment, the container is capable of maintaining the formed configuration (e.g.,

FIGS. **1**F and **2**F) once the container is moved from the box configuration (e.g., FIGS. **1**E and **2**E) to the formed configuration (e.g., FIGS. **1**F and **2**F). This is enabled by the structure of the container. The container on its own may maintain the formed configuration without using an added mechanism (e.g., mechanical interlock, fastening, etc.) and without using an external force.
(41) In an embodiment, the separation panel **103**, **203** extends beyond the first side panel **101**, **201**, beyond the second side panel **111**, **211**, or beyond both when the container is in the formed configuration (e.g., FIGS. **1**F and **2**F). For example, this may allow objects such as cups, beverages, bottles, etc. to be held and/or carried in the container.
(42) In an embodiment, the separation panel **103**, **203** is configured to compartmentalize a usable space of the container when the container is in the formed configuration (e.g., FIGS. **1**F and **2**F). For example, a compartmentalized container divided into multiple sections may allow multiple objects such as cups, beverages, bottles, etc. to be held and/or carried in the container.
(43) In an embodiment, the container further comprises at least one end flap **115**, **125**, **215**, **225** hingedly coupled to the first side panel **101**, **201**. The at least one end flap **115**, **125**, **215**, **225** and the separation panel **103**, **203** are substantially planar with respect to each other when the container is in the box configuration (e.g., FIGS. **1**E and **2**E). The at least one end flap **115**, **125**, **215**, **225** and the separation panel **103**, **203** are substantially orthogonal with respect to each other when the container is in the formed configuration (e.g., FIGS. **1**F and **2**F). For example, the end flaps **115**, **125**, **215**, **225** may be used to secure objects such as cups, beverages, bottles, etc. and to prevent them from falling from the container. The end flaps **115**, **125**, **215**, **225** may be hingedly coupled to end tabs **116**, **126**, **216**, **226** (e.g., FIGS. **1**A and **2**A). The end tabs **116**, **126** and a complementary side panel **121** may be joined or attached to each other (e.g., FIGS. **1**E and **1**F). Alternatively, the end tabs **216**, **226** and the second side panel **211** may be joined or attached to each other (e.g., FIG. **2**C). As another alternative variation, the at least one end flap may be hingedly coupled to the second side panel prior to folding a blank.
(44) In an embodiment, the container is made from a blank **100**, **200**, **300***a*, **300***b*, **300***c*, **300***d*, **300***e*, **300***f*, **400***a*, **400***b*, **400***c*, **400***d*, **400***e*, **400***f*, **500***a*, **500***b* (e.g., FIGS. **1**A, **2**A, **3**A-**3**F, **4**A-**4**F, and **5**A-**5**B).
(45) In an embodiment, the first side panel **101**, **201** is hingedly coupled to the bottom panel **107**, **207** (e.g., FIGS. **1**A and **2**A).
(46) In an embodiment, the second side panel **111**, **211** is hingedly coupled to the bottom panel **107**, **207** (e.g., FIGS. **1**E and **2**E). The coupling between the second side panel **111**, **211** and the bottom panel **107**, **207** may be achieved in various ways. For example, the second side panel **311** may be hingedly coupled to the bottom panel **307** prior to folding a blank (e.g., FIG. **3**F). Alternatively, this coupling may be achieved using a complementary side panel **121**, **321**, which is hingedly coupled to the bottom panel **107**, **307** (e.g., FIGS. **1**A and **3**E). The complementary side panel **121**, **321** and the second side panel **111**, **311** may be joined or attached to each other (e.g., FIGS. **1**D and **1**E). Furthermore, this coupling may be achieved using a base portion **220**, which is hingedly coupled to the second side panel **211** (e.g., FIG. **2**A). The base portion **220** and the bottom panel **207** may be joined or attached to each other (e.g., FIGS. **2**B and **2**E).
(47) In an embodiment, the first side portion **502** comprises at least one additional hinge **551** to facilitate moving the container from the box configuration to the formed configuration (e.g., FIG. **5**A). This may provide a better user experience when moving the container from the box configuration to the formed configuration.
(48) In an embodiment, the second side portion **512** comprises at least one additional hinge **552** to facilitate moving the container from the box configuration to the formed configuration (e.g., FIG. **5**A). This may provide a better user experience when moving the container from the box configuration to the formed configuration.
(49) In an embodiment, the separation panel **503** further comprises at least one additional hinge **553** to facilitate moving the container from the box configuration to the formed configuration (e.g.,

FIG. **5**A). This may provide a better user experience when moving the container from the box configuration to the formed configuration.
(50) In an embodiment, the separation panel **203** is a first separation panel **203**. The container further comprises a third side panel **221**, a fourth side panel **231** and a second separation panel **213**. The third side panel **221** is hingedly coupled to a third side portion **222**. The fourth side panel **231** is hingedly coupled to a fourth side portion **232**. The second separation panel **213** is hingedly coupled to the third side portion **222**. The second separation panel **213** is hingedly coupled to the fourth side portion **232** (e.g., FIG. **2**A).
(51) In an embodiment, the first separation panel **203** and the second separation panel **213** are joined to each other (e.g., FIGS. **2**D, **2**E, and **2**F). The first separation panel **203** and the second separation panel **213** may be joined or attached to each other using glue, adhesives, polymers, chemicals, etc.
(52) In an embodiment, the fourth side panel **231** and the second side panel **211** are joined to each other (e.g., FIGS. **2**D, **2**E, and **2**F; the fourth side panel **231** is not visible). The fourth side panel **231** and the second side panel **211** may be joined or attached to each other using glue, adhesives, polymers, chemicals, etc.
(53) In an embodiment, the separation panel **503** is hingedly coupled to a partial separation panel **560** (e.g., FIG. **5**B). The separation panel **503** and the partial separation panel **560** may be joined or attached to each other using glue, adhesives, polymers, chemicals, etc. The partial separation panel **560** may reinforce the separation panel **503**.
(54) With reference to FIGS. **1**D-**1**F and **2**D-**2**F, embodiments are directed to a container movable between a flat configuration (e.g., FIGS. **1**D and **2**D) and a box configuration (e.g., FIGS. **1**E and **2**E). The container comprises a first side panel **101**, **201** hingedly coupled to a first side portion **102**, **202**, a second side panel **111**, **211** hingedly coupled to a second side portion **112**, **212**, a separation panel **103**, **203**, and a bottom panel **107**, **207**. The separation panel **103**, **203** is hingedly coupled to the first side portion **102**, **202**. The separation panel **103**, **203** is hingedly coupled to the second side portion **112**, **212**. The bottom panel **107**, **207** and the separation panel **103**, **203** face each other when the container is in the box configuration (e.g., FIGS. **1**E and **2**E). The first side panel **101**, **201** and the second side panel **111**, **211** face each other when the container is in the box configuration (e.g., FIGS. **1**E and **2**E). The container is further movable between the box configuration (e.g., FIGS. **1**E and **2**E) and a formed configuration (e.g., FIGS. **1**F and **2**F). The first side panel **101**, **201** and the second side panel **111**, **211** face each other when the container is in the formed configuration (e.g., FIGS. **1**F and **2**F). The first side portion **102**, **202** is configured to fold towards the first side panel **101**, **201**. The second side portion **112**, **212** is configured to fold towards the second side panel **111**, **211**. The first side portion **102**, **202** and the first side panel **101**, **201** face each other when the container is in the formed configuration (e.g., FIGS. **1**F and **2**F; the first side portion **102**, **202** is not visible). The second side portion **112**, **212** and the second side panel **111**, **211** face each other when the container is in the formed configuration (e.g., FIGS. **1**F and **2**F). The separation panel **103**, **203** and the bottom panel **107**, **207** are substantially orthogonal with respect to each other when the container is in the formed configuration (e.g., FIGS. **1**F and **2**F).
(55) In an embodiment, the container is made from a blank **100**, **200**, **300***a*, **300***b*, **300***c*, **300***d*, **300***e*, **300***f*, **400***a*, **400***b*, **400***c*, **400***d*, **400***e*, **400***f*, **500***a*, **500***b* (e.g., FIGS. **1**A, **2**A, **3**A-**3**F, **4**A-**4**F, and **5**A-**5**B).
(56) Although embodiments are described above with reference to a container or a blank used to make the container, wherein the at least one end flap is shown hingedly coupled to the first side panel prior to folding the blank, the at least one end flap described in any of the above embodiments may alternatively be configured to be hingedly coupled to the second side panel prior to folding the blank. Such alternatives are considered to be within the spirit and scope of the present disclosure, and may therefore utilize the advantages of the configurations and embodiments

described above.
(57) In addition, although embodiments are described above with reference to a container, wherein the first side portion and the first side panel are shown hingedly coupled via a 45° hinge, the structure and design features of the first side portion, the first side panel, and the hinge coupling them described in any of the above embodiments may vary based on the desired objectives and applications. Portions or all of the container may be configured to accommodate such alternatives. Such alternatives are considered to be within the spirit and scope of the present disclosure, and may therefore utilize the advantages of the configurations and embodiments described above.
(58) Further, although embodiments are described above with reference to a container, wherein the separation panel and the bottom panel are shown perpendicular with each other when the container is in the formed configuration, the angle between the separation panel and the bottom panel described in any of the above embodiments may vary. Portions or all of the container may be configured to accommodate such alternatives. Such alternatives are considered to be within the spirit and scope of the present disclosure, and may therefore utilize the advantages of the configurations and embodiments described above.
(59) Yet further, although embodiments are described above with reference to a container, wherein the container is shown with uniform portions (e.g., with uniform thickness), portions or all of the container described in any of the above embodiments may alternatively be non-uniform (e.g., having varying thickness, discontinuities such as holes, slits, grooves, ridges, slits, a combination thereof, etc.). Such alternatives are considered to be within the spirit and scope of the present disclosure, and may therefore utilize the advantages of the configurations and embodiments described above.
(60) The method steps in any of the embodiments described herein are not restricted to being performed in any particular order. Also, structures mentioned in any of the method embodiments may utilize structures mentioned in any of the device embodiments. Such structures may be described in detail with respect to the device embodiments only but are applicable to any of the method embodiments.
(61) Features in any of the embodiments described in this disclosure may be employed in combination with features in other embodiments described herein, such combinations are considered to be within the spirit and scope of the present disclosure.
(62) The contemplated modifications and variations specifically mentioned in this disclosure are considered to be within the spirit and scope of the present disclosure.
(63) More generally, even though the present disclosure and exemplary embodiments are described above with reference to the examples according to the accompanying drawings, it is to be understood that they are not restricted thereto. Rather, it is apparent to those skilled in the art that the disclosed embodiments can be modified in many ways without departing from the scope of the disclosure herein. Moreover, the terms and descriptions used herein are set forth by way of illustration only and are not meant as limitations. Those skilled in the art will recognize that many variations are possible within the spirit and scope of the disclosure as defined in the following claims, and their equivalents, in which all terms are to be understood in their broadest possible sense unless otherwise indicated.
(64) For the figures referred to herein, a summary of the reference numbers used in the figures is included in Table 1, below.
(65) TABLE-US-00001 TABLE 1 Reference Index for Drawings Description FIGS. 1A-1F FIGS. 2A-2F FIGS. 3A-3F FIGS. 4A-4F FIGS. 5A-5B Blank 100 200 300a, 300b, 400a, 400b, 500a, 500b 300c, 300d, 400c, 400d, 300e, 300f 400e, 400f First side panel 101 201 501 First side portion 102 202 502 Second side panel 111 211 311 411 511 Second side 112 212 512 portion Separation panel 103 203 503 (first) Handle 104 204 Bottom panel 107 207 307 507 End flap 115, 125 215, 225 End tab 116, 126 216, 226 Complementary 121 321 side panel Base portion 220 Third side panel 221 Third side portion 222 Fourth side panel 231 Fourth side 232 portion Second separation

213 panel Additional hinge 551, 552, 553 Partial separation 560 panel
(66) Those skilled in the art will recognize or be able to ascertain using no more than routine experimentation, many equivalents to the specific embodiments of the disclosure described herein. Such equivalents are intended to be encompassed by the following claims.

Claims

1. A container movable between a flat configuration and a box configuration, the container comprising: a first side panel, wherein the first side panel is hingedly coupled to a first side portion; a second side panel, wherein the second side panel is hingedly coupled to a second side portion; a separation panel, wherein the separation panel is hingedly coupled to the first side portion, wherein the separation panel is hingedly coupled to the second side portion, and wherein the separation panel comprises a handle; a bottom panel, wherein the bottom panel and the separation panel face each other when the container is in the box configuration; wherein the first side panel and the second side panel face each other when the container is in the box configuration; wherein the container is further movable between the box configuration and a formed configuration; wherein the first side portion is configured to fold towards the first side panel, wherein the second side portion is configured to fold towards the second side panel, wherein the first side portion and the first side panel face each other when the container is in the formed configuration, wherein the second side portion and the second side panel face each other when the container is in the formed configuration, wherein the separation panel and the bottom panel are substantially orthogonal with respect to each other when the container is in the formed configuration; wherein the container is configured to be held via the handle when the container is in the formed configuration; and wherein the separation panel is configured to compartmentalize a usable space of the container when the container is in the formed configuration.
2. The container of claim 1, wherein the container is capable of maintaining the formed configuration once the container is moved from the box configuration to the formed configuration.
3. The container of claim 1, wherein the separation panel extends beyond the first side panel, beyond the second side panel, or beyond both when the container is in the formed configuration.
4. The container of claim 1, wherein the container further comprises at least one end flap, wherein the at least one end flap is hingedly coupled to the first side panel, wherein the at least one end flap and the separation panel are substantially planar with respect to each other when the container is in the box configuration, and wherein the at least one end flap and the separation panel are substantially orthogonal with respect to each other when the container is in the formed configuration.
5. The container of claim 1, wherein the container is made from a blank.
6. The container of claim 1, wherein the first side panel is hingedly coupled to the bottom panel.
7. The container of claim 1, wherein the second side panel is hingedly coupled to the bottom panel.
8. The container of claim 1, wherein the first side portion comprises at least one additional hinge to facilitate moving the container from the box configuration to the formed configuration.
9. The container of claim 1, wherein the second side portion comprises at least one additional hinge to facilitate moving the container from the box configuration to the formed configuration.
10. The container of claim 1, wherein the separation panel further comprises at least one additional hinge to facilitate moving the container from the box configuration to the formed configuration.
11. The container of claim 1, wherein the separation panel is a first separation panel, wherein the container further comprises a third side panel, a fourth side panel and a second separation panel, wherein the third side panel is hingedly coupled to a third side portion, wherein the fourth side panel is hingedly coupled to a fourth side portion, wherein the second separation panel is hingedly coupled to the third side portion, and wherein the second separation panel is hingedly coupled to the fourth side portion.

12. The container of claim 11, wherein the first separation panel and the second separation panel are joined to each other.
13. The container of claim 11, wherein the fourth side panel and the second side panel are joined to each other.
14. A container movable between a flat configuration and a box configuration, the container comprising: a first side panel, wherein the first side panel is hingedly coupled to a first side portion; a second side panel, wherein the second side panel is hingedly coupled to a second side portion; a separation panel, wherein the separation panel is hingedly coupled to the first side portion, and wherein the separation panel is hingedly coupled to the second side portion; a bottom panel, wherein the bottom panel and the separation panel face each other when the container is in the box configuration; wherein the first side panel and the second side panel face each other when the container is in the box configuration; wherein the container is further movable between the box configuration and a formed configuration; wherein the first side portion is configured to fold towards the first side panel, wherein the second side portion is configured to fold towards the second side panel, wherein the first side portion and the first side panel face each other when the container is in the formed configuration, wherein the second side portion and the second side panel face each other when the container is in the formed configuration, wherein the separation panel and the bottom panel are substantially orthogonal with respect to each other when the container is in the formed configuration; and wherein the separation panel is configured to compartmentalize a usable space of the container when the container is in the formed configuration.
15. The container of claim 14, wherein the container is made from a blank.